

RESULTS

# Results

01 · DIFFERENCE-IN-DIFFERENCES (DID)

policy effect, before/after × treated/control

**Table.** DiD model

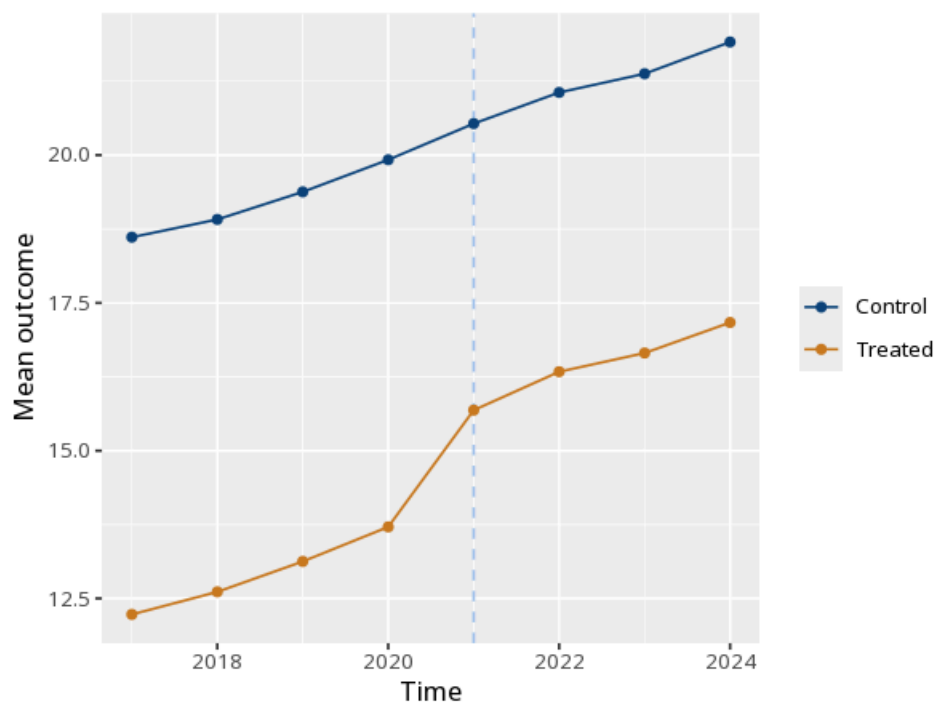
|                | (1)                            |
|----------------|--------------------------------|
| Post           | 2.02<br>(0.09)<br>[1.84, 2.19] |
| Treated × Post | 1.53<br>(0.12)<br>[1.29, 1.76] |
| Num.Obs.       | 96                             |
| N entities     | 12                             |
| Within R²      | .84                            |
| F              | F(2, 82) = 216.31, p < .001    |

**Table 2.** Group × period means

| Group   | Pre   | Post  |
|---------|-------|-------|
| Control | 19.21 | 21.22 |
| Treated | 12.92 | 16.46 |

the Treated×Post interaction is the DiD effect; valid only if pre-treatment trends are parallel. Under entity fixed effects the time-invariant Treated main effect is absorbed (only Post and Treated×Post are estimated). Standard errors in parentheses are clustered by entity; the bracketed line is the 95% CI. Pre-trends test (pre-period leads-and-lags joint F of treated×time): F(3, 40) = 0.00, p = 1.000 — a small p flags diverging pre-trends.

**Figure.** Parallel trends



## Understanding the numbers

**Treated×Post (B).** The estimated causal effect of the treatment, holding entity and period fixed effects constant. Here,  $B = 1.53$ , 95% CI [1.29, 1.76],  $p < .001$ .

**N.** The number of observations (entity-period rows) used to fit the model. Here,  $N = 96$ .

**N entities.** How many distinct units (e.g. states, firms) are represented, each contributing its own fixed effect. Here,  $N_{\text{entities}} = 12$ .

**Within  $R^2$ .** The share of within-entity variance (after removing each entity's own average) explained by the model. Here,  $\text{within } R^2 = .84$ .

**F.** The overall model F-test for the within regression. Here, the overall model test is  $F(2, 82) = 216.31$ ,  $p < .001$ .

**Pre.** The raw (unadjusted) mean outcome for this group before treatment. Here, the Control group's pre-period mean is 19.21.

**Post.** The raw (unadjusted) mean outcome for this group after treatment. Here, the Control group's post-period mean is 21.22.

## How to read this test

The Treated×Post coefficient is the estimated treatment effect. It rests on the parallel-trends assumption: that the groups would have moved together absent treatment. Similar pre-treatment trends (inspect the pre-period of the plot) make this more plausible but do not prove it — a visual check is supportive, not confirmatory, since the assumption is about the unobservable post-period counterfactual. The note adds a formal pre-trends test (a pre-period leads-and-lags joint F of the treated×time interactions): a small  $p$  flags diverging pre-trends, while a large  $p$  is consistent with parallel trends.

Method: R plm package with clustered standard errors (Croissant & Millo, 2008; Bertrand, Duflo & Mullainathan, 2004).  
Your significance threshold ( $\alpha$ ) is 0.05.

**APA template:** The DiD estimate was  $B=1.53$ , 95% CI [1.29, 1.76],  $p < .001$  (clustered SE).

**Statistical basis:** Difference-in-differences via entity fixed effects (plm within estimator) (Croissant, Y., Millo, G. (2008). "Panel Data Econometrics in R: The plm Package." Journal of Statistical Software, 27(2), 1-43.); Clustered standard errors / parallel-trends interpretation (Bertrand, M., Duflo, E., Mullainathan, S. (2004). "How Much Should We Trust Differences-in-Differences Estimates?" Quarterly Journal of Economics, 119(1), 249-275.). Full references in the exported CITATIONS.txt.

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